AWS CLOUD · STEP-BY-STEP GUIDE

Connecting Amazon RDS with Amazon EC2

Create a MySQL database, launch an Ubuntu server and connect them securely, illustrated with real AWS Console screenshots.

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Region: Asia Pacific (Sydney) ap-southeast-2

16 October 2026

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PART 1

Create the RDS database

Steps 1–8

1 Open the Amazon RDS dashboard

Sign in to the AWS Console, select the region (here Asia Pacific (Sydney), ap-southeast-2) and open Aurora and RDS from the service shortcuts.

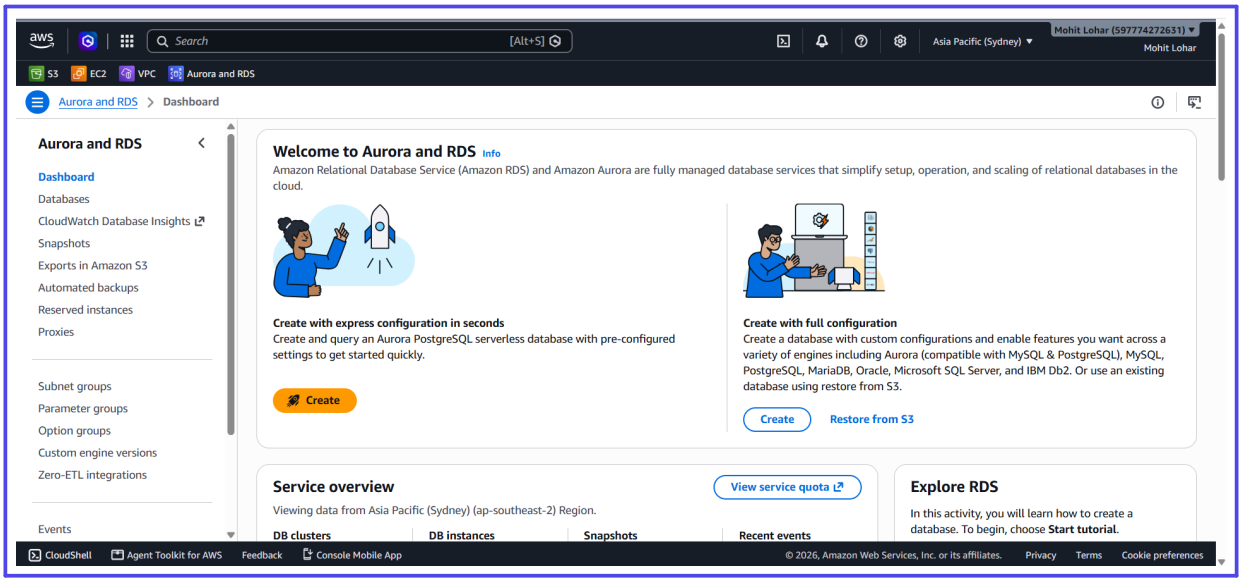


Figure 1 · Open the Amazon RDS dashboard

NOTES

- Make sure the region is the same one you will use for EC2 - RDS and EC2 must be in the same region and VPC.
- Choose Databases in the left menu to see existing databases.

2 Review existing databases

The Databases page lists what already exists in the region. Here an Aurora PostgreSQL cluster named database-1 (with writer instance database-1-instance-1) is shown as Stopped temporarily.

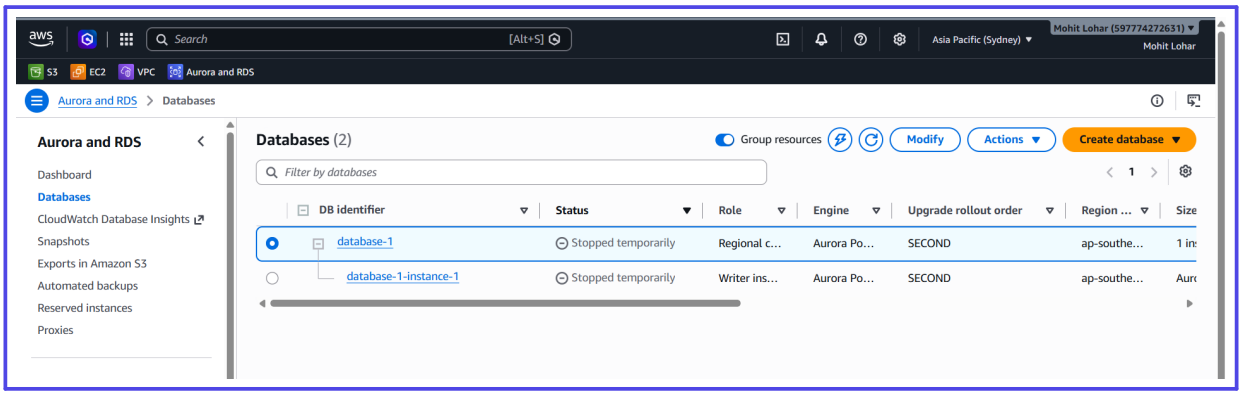


Figure 2 · Review existing databases

NOTES

- This is separate from the MySQL instance we are about to create.
- Click Create database.

3 Choose Full configuration

From the Create database drop-down select Full configuration so every option (engine, template, credentials, storage) can be set manually.

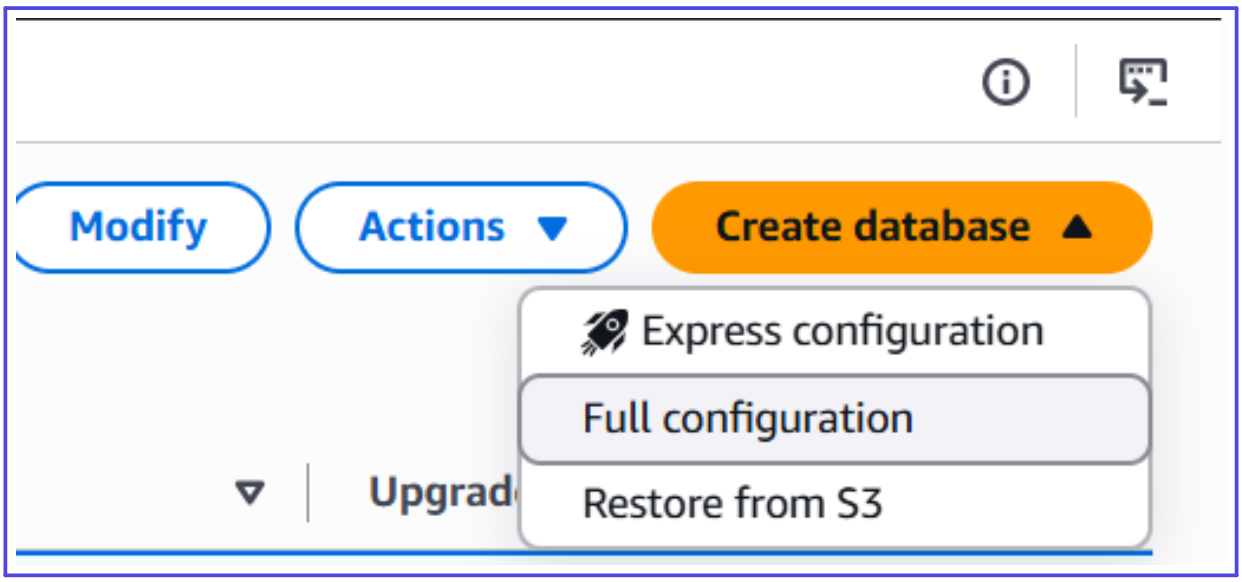


Figure 3 · Choose Full configuration

NOTES

- Express configuration creates an Aurora PostgreSQL serverless database - not what we need here.

4 Select the database engine

Under Engine options select MySQL. The edition is MySQL Community.

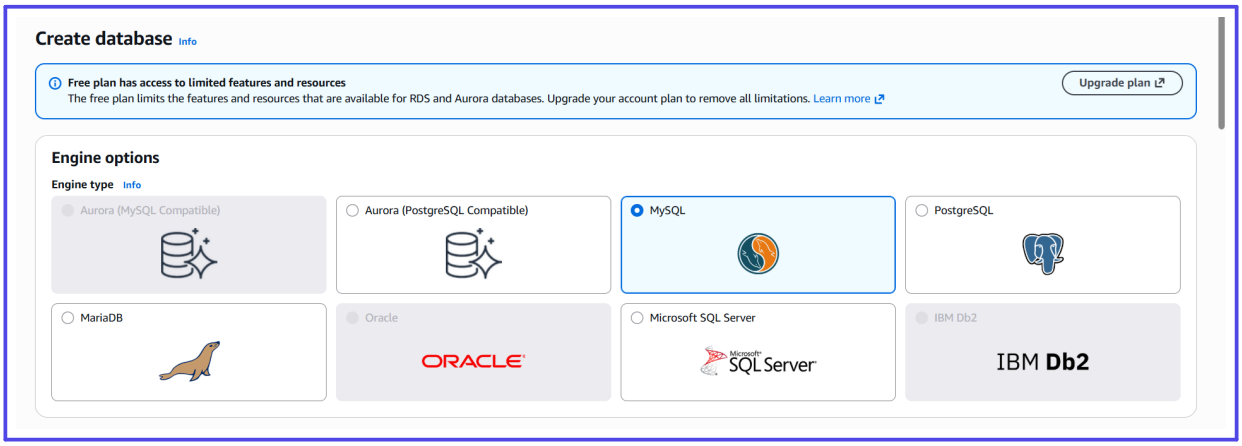


Figure 4 · Select the database engine

NOTES

- A banner notes the Free plan limits some features; greyed-out engines are not available on the free plan.

5 Choose creation method and template

Keep Full configuration as the creation method and pick the Free tier template. Production and Dev/Test are disabled on the free plan.

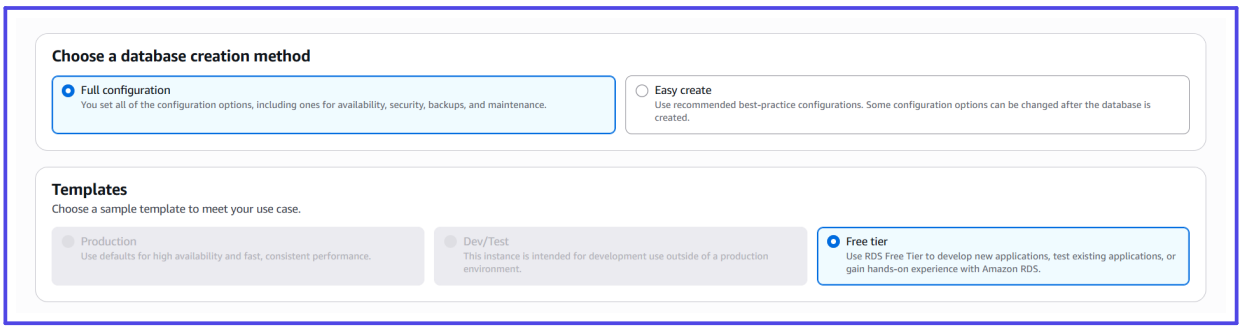


Figure 5 · Choose creation method and template

NOTES

- Free tier keeps cost at zero within the allowed limits (small instance class, limited storage).

6 Set identifier, username and credentials management

Configure the basic settings:

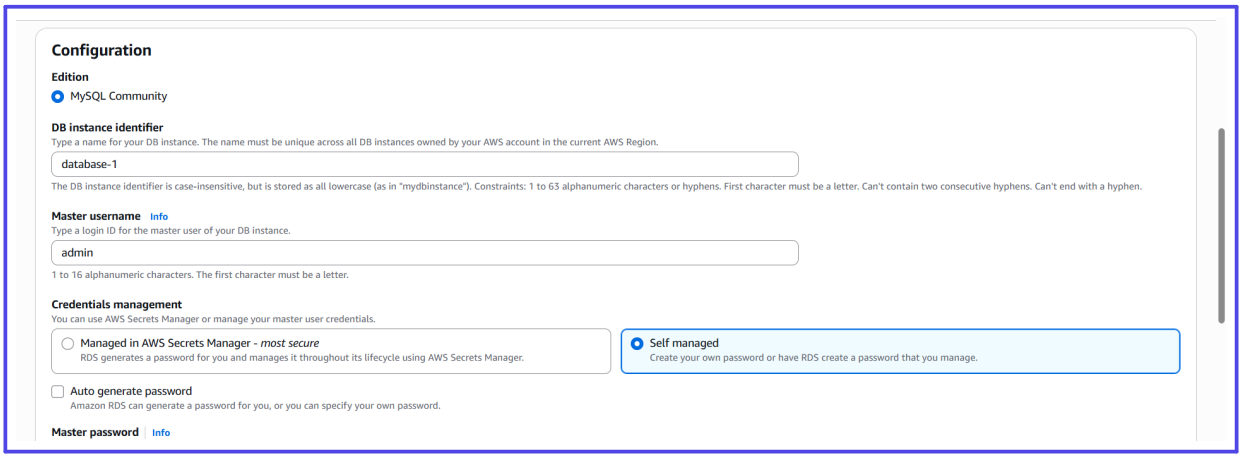


Figure 6 · Set identifier, username and credentials management

NOTES

- DB instance identifier: database-1
- Master username: admin
- Credentials management: Self managed (you choose the password). Managed in AWS Secrets Manager is the more secure option for real projects.
- Leave Auto generate password unticked and type your own password next.

7 Set the master password and authentication

Enter and confirm the master password. The console shows password strength (Average here). Database authentication is left as Password authentication.

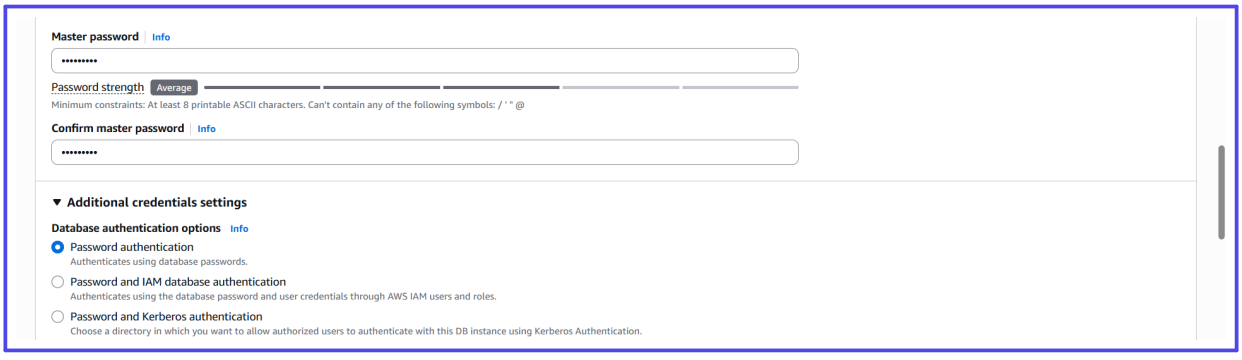


Figure 7 · Set the master password and authentication

NOTES

- Password rules: at least 8 printable ASCII characters; cannot contain / ' " or @.
- Use a stronger, unique password and store it safely - it is needed later to log in from EC2.

8 Configure storage and create the database

Storage type is General Purpose SSD (gp2) with 20 GiB allocated (the allowed range is 20 to 6,144 GiB). Review the remaining sections, then click Create database.

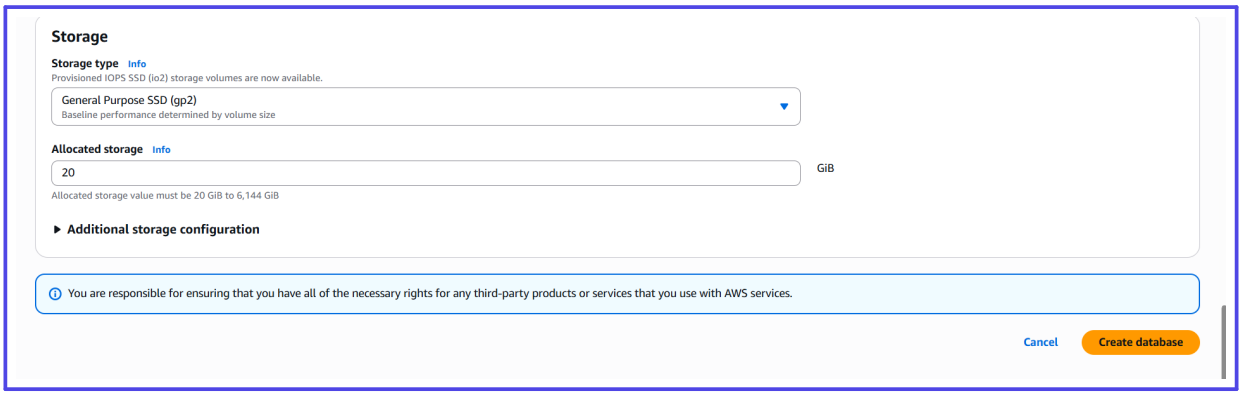


Figure 8 · Configure storage and create the database

NOTES

- Wait until the status changes from Creating to Available before connecting.

PART 2

Launch the EC2 instance

Steps 9–16

9 Open the EC2 console

Open EC2 from the service shortcuts and click Launch instance.

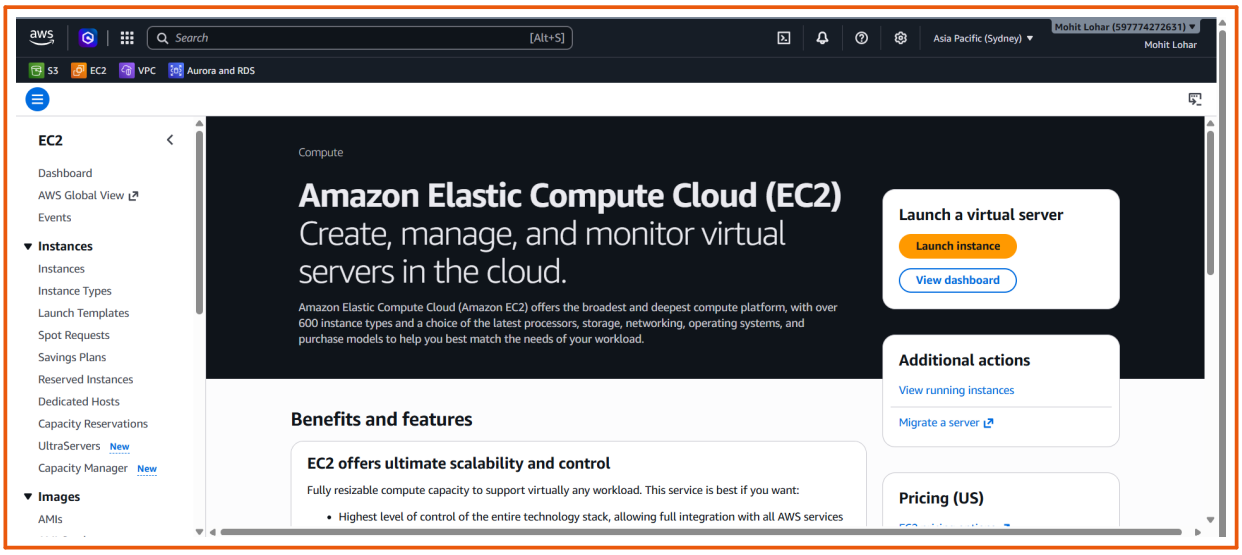


Figure 9 · Open the EC2 console

NOTES

- Use the same region as the database (Sydney).

10 Name the instance

Under Name and tags enter the instance name: EC2-RDS.

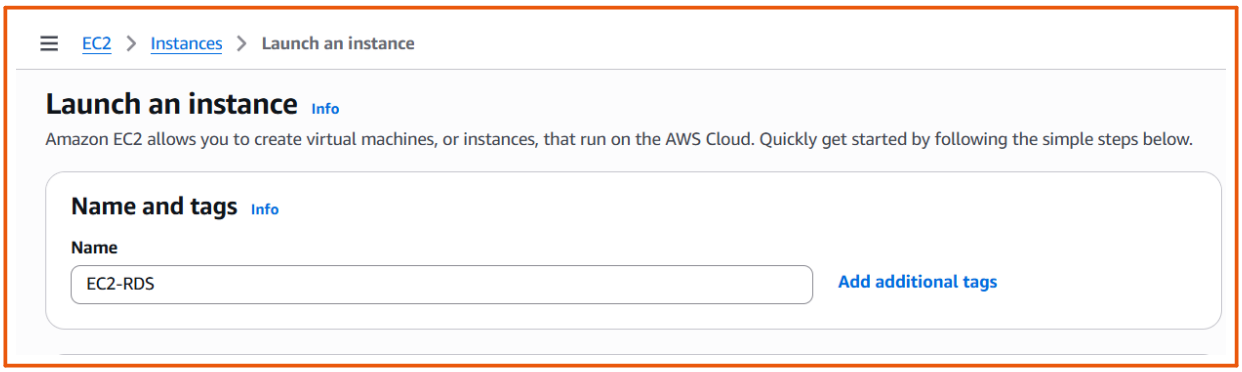


Figure 10 · Name the instance

11 Choose the operating system (AMI)

From Quick Start select Ubuntu. The AMI shown is Ubuntu Server 26.04 LTS (HVM), SSD Volume Type, marked Free tier eligible.

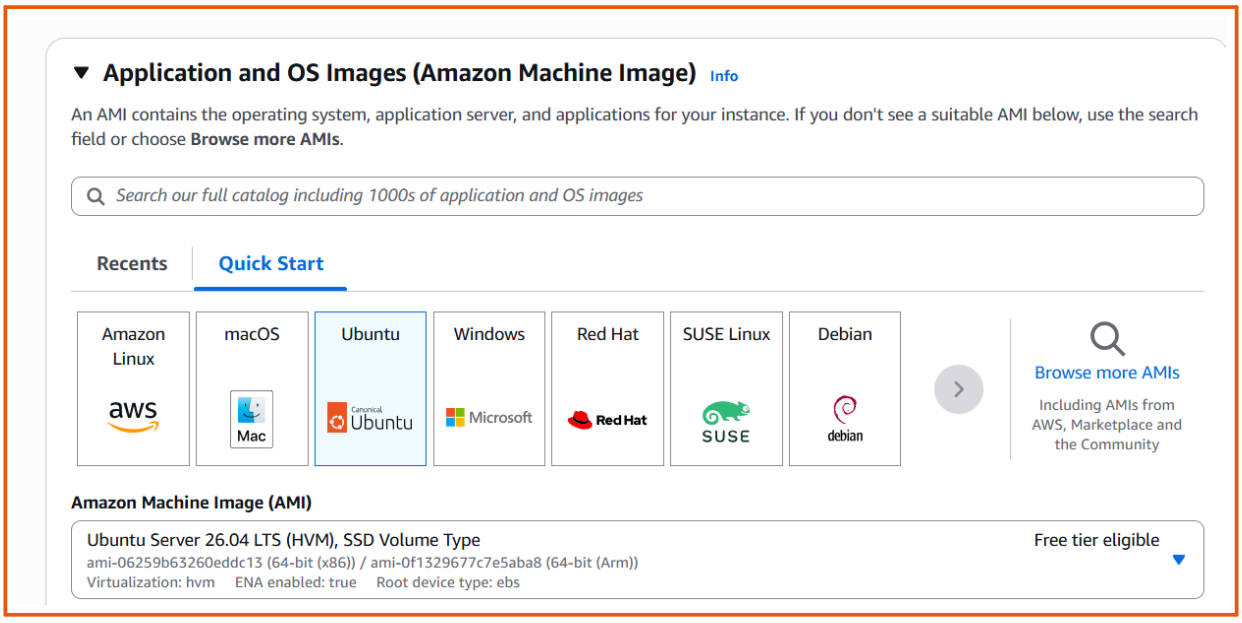


Figure 11 · Choose the operating system (AMI)

12 Choose instance type and key pair

Instance type: t3.micro (2 vCPU, 1 GiB memory, free tier eligible). Key pair: Proceed without a key pair was selected.

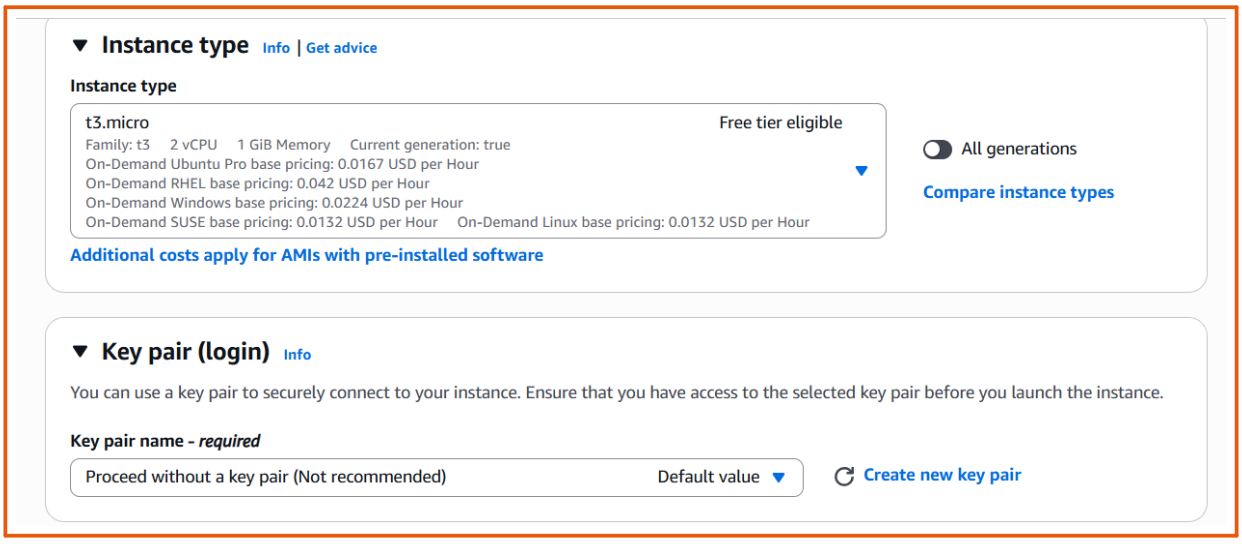


Figure 12 · Choose instance type and key pair

ATTENTION

- Without a key pair you cannot SSH from your own computer; you can still connect using the browser-based EC2 Instance Connect / CloudShell option.
- Recommended: create a key pair (.pem) so you can SSH from a terminal.

13 Configure network settings and security group

Network: default VPC vpc-0660db764b25a72f1, subnet No preference, Auto-assign public IP enabled. A new security group launch-wizard-2 is created allowing SSH, HTTPS and HTTP.

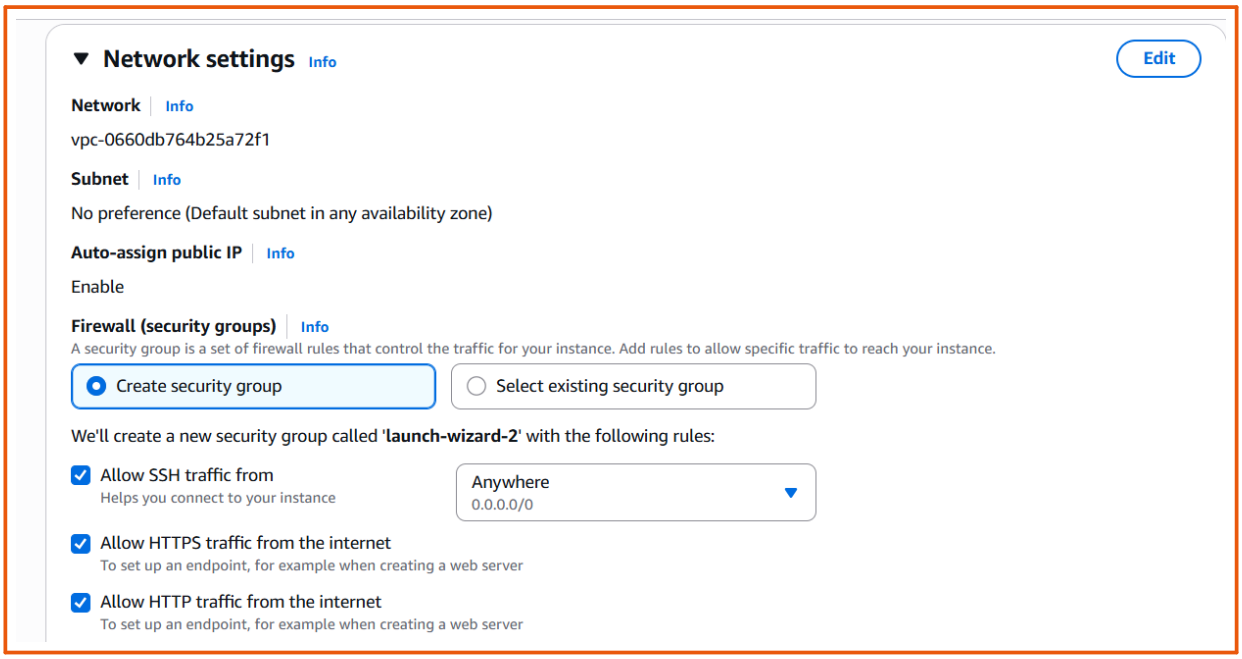


Figure 13 · Configure network settings and security group

NOTES

- Only open HTTP/HTTPS if the instance will serve a web application.

ATTENTION

- SSH is open to Anywhere (0.0.0.0/0). For anything beyond a lab, change it to My IP.

14 Review the summary and launch

Summary: 1 instance, Ubuntu 26.04 AMI, t3.micro, new security group, 1 volume of 8 GiB. Click Launch instance.

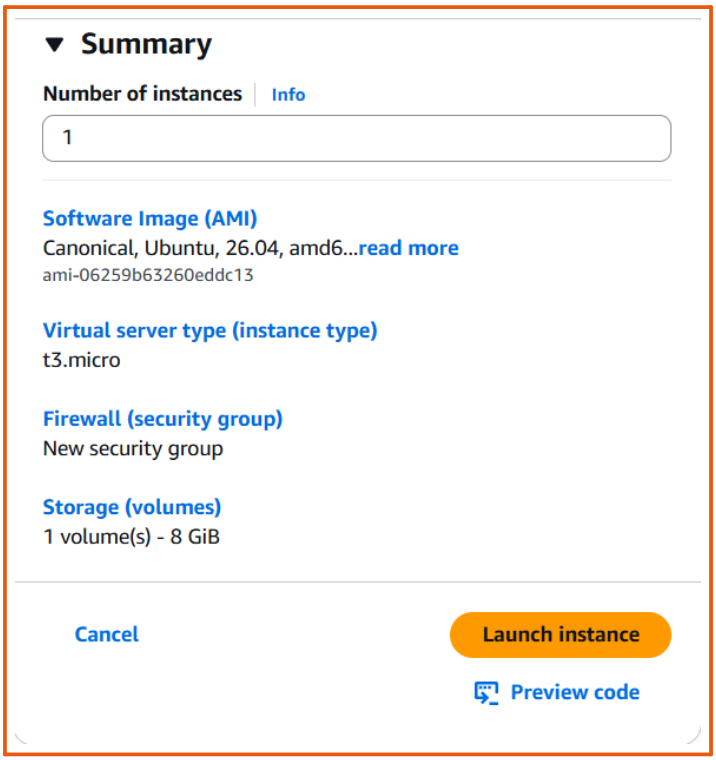


Figure 14 · Review the summary and launch

15 Confirm the instance is running

The instance EC2-RDS (i-05f2f8969b84312d5) shows state Running in ap-southeast-2b; status checks are still Initializing. Wait until they pass, then click Connect.

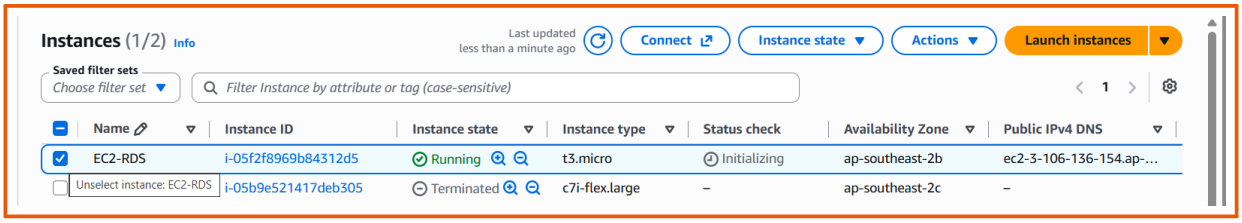


Figure 15 · Confirm the instance is running

NOTES

- A previously terminated instance (c7i-flex.large) is also listed - it can be ignored.

16 Connect to the instance terminal

The browser terminal opens an Ubuntu shell (private IP 172.31.41.2). Update the package list before installing anything:

```
$ sudo apt update -y
```

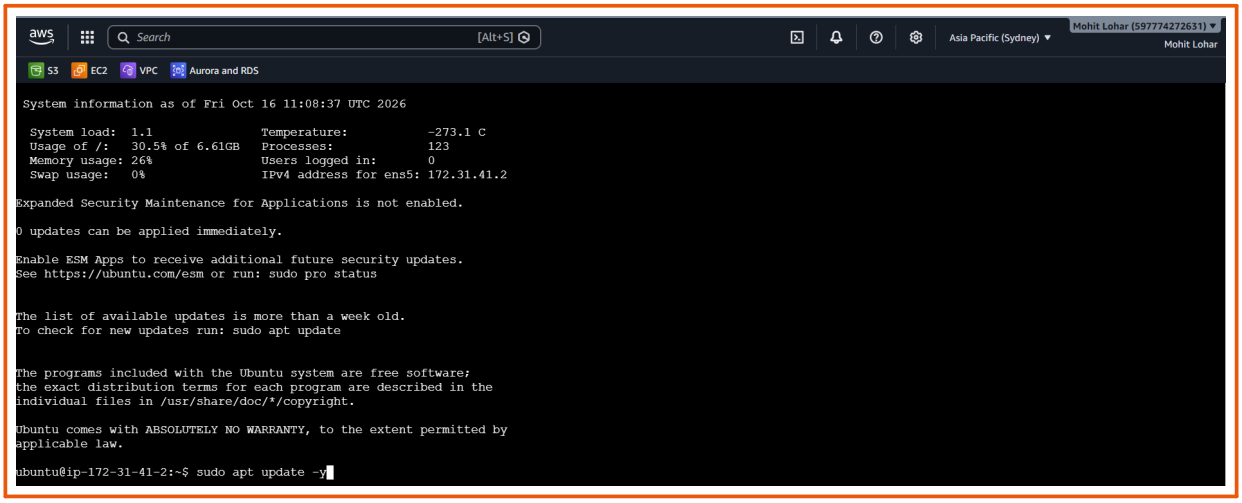


Figure 16 · Connect to the instance terminal

PART 3 Steps 17–22

Connect EC2 to RDS

17 Verify the MySQL database is Available

Back in RDS > Databases, the new MySQL Community instance database-1 shows status Available. The two Aurora PostgreSQL entries above it are the older stopped cluster.

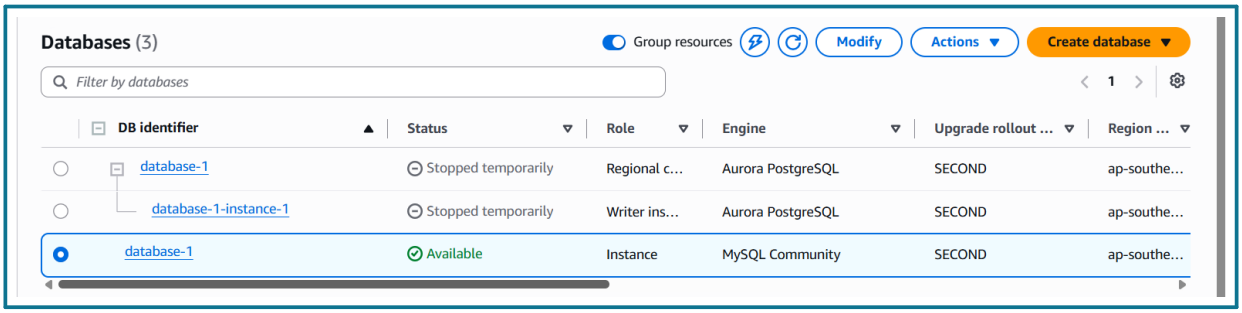


Figure 17 · Verify the MySQL database is Available

NOTES

- Click the MySQL database-1 instance to open its details.

18 Open Connected compute resources

On the database details page scroll to Connected compute resources. It shows 0 resources. Click Set up EC2 connection.

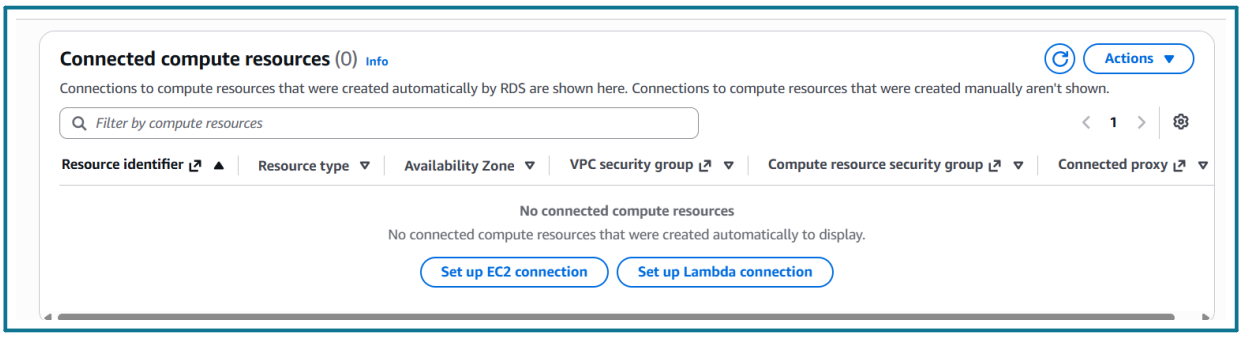


Figure 18 · Open Connected compute resources

19 Select the EC2 instance

Choose i-05f2f8969b84312d5 (EC2-RDS, ap-southeast-2b). Only EC2 instances in the same VPC as the database are listed. Click Continue.

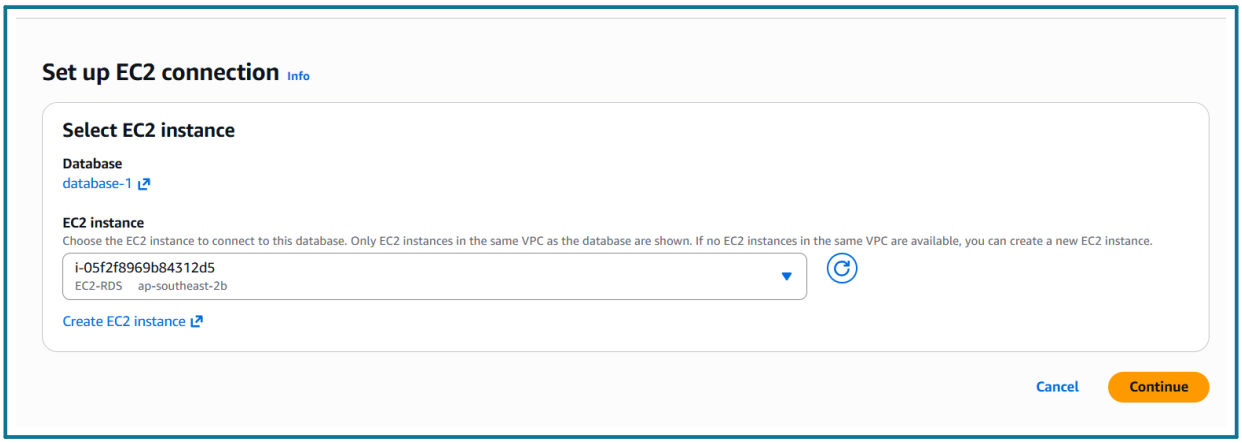


Figure 19 · Select the EC2 instance

20 Review the connection summary

RDS shows what it will configure automatically: security group rds-ec2-1 is added to the database and security group ec2-rds-1 is added to the EC2 instance, allowing traffic on port 3306 between them.

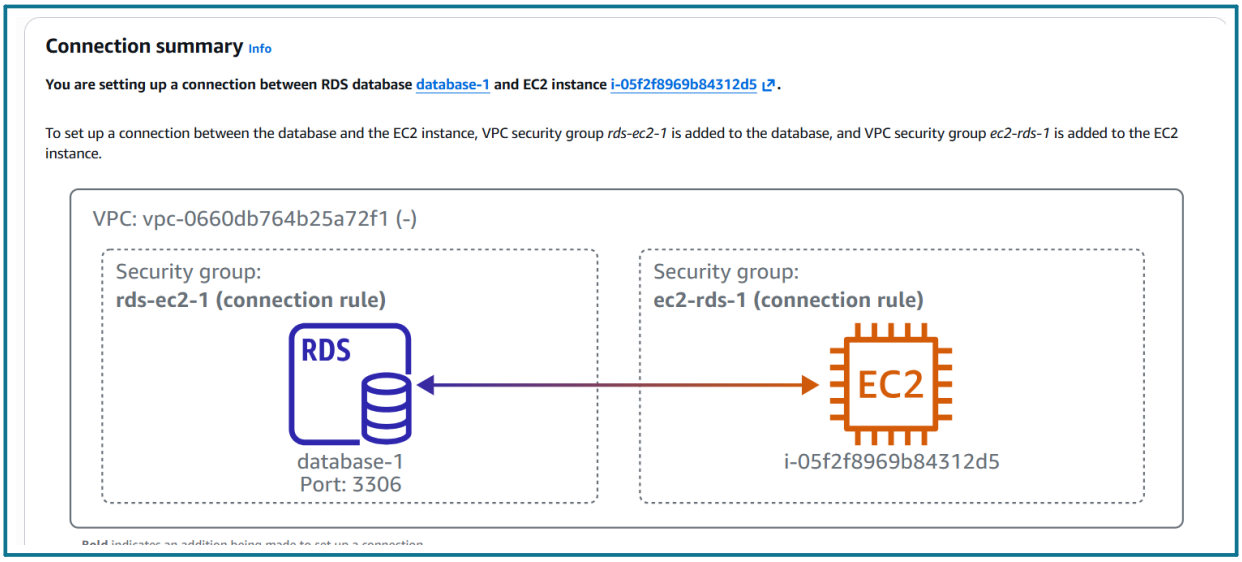


Figure 20 · Review the connection summary

NOTES

- Scroll down on the same page to review the exact changes (next step).
- This is more secure than opening port 3306 to the internet - only this EC2 instance can reach the database.

21 Review the changes and click Set up

Scrolling down shows the exact changes RDS will make. The database keeps the default security group and gets rds-ec2-1 added. The EC2 instance keeps launch-wizard-2 and gets ec2-rds-1 added. Click Set up.

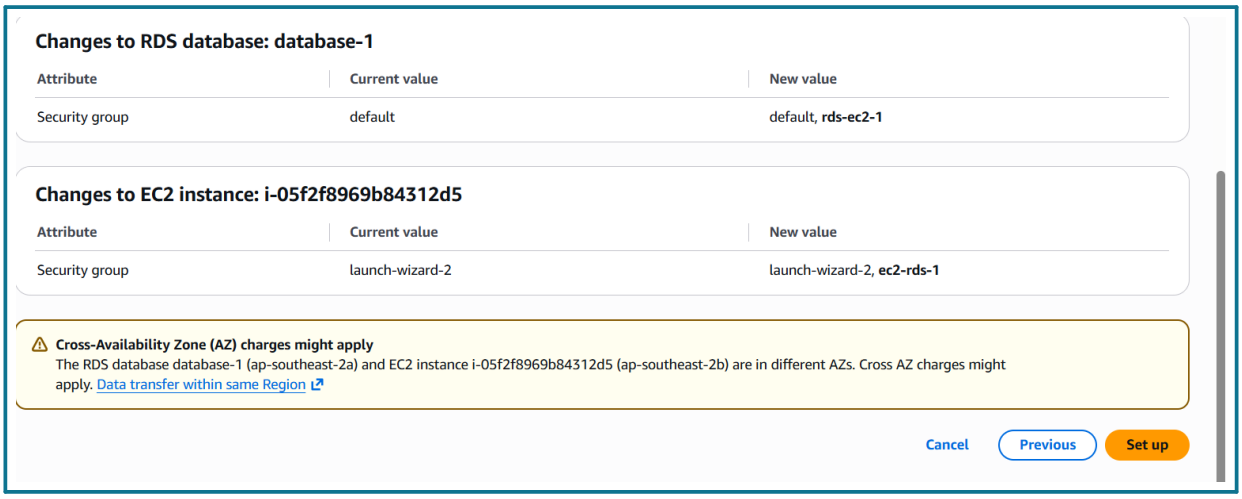


Figure 21 · Review the changes and click Set up

ATTENTION

- Warning shown: the database is in ap-southeast-2a and the EC2 instance is in ap-southeast-2b, so cross-AZ data transfer charges might apply. To avoid this, launch the EC2 instance in the same Availability Zone as the database.

22 Connection set up successfully

A green banner confirms: Connection setup successfully for RDS database database-1 and EC2 instance i-05f2f8969b84312d5. The MySQL database-1 shows status Available.

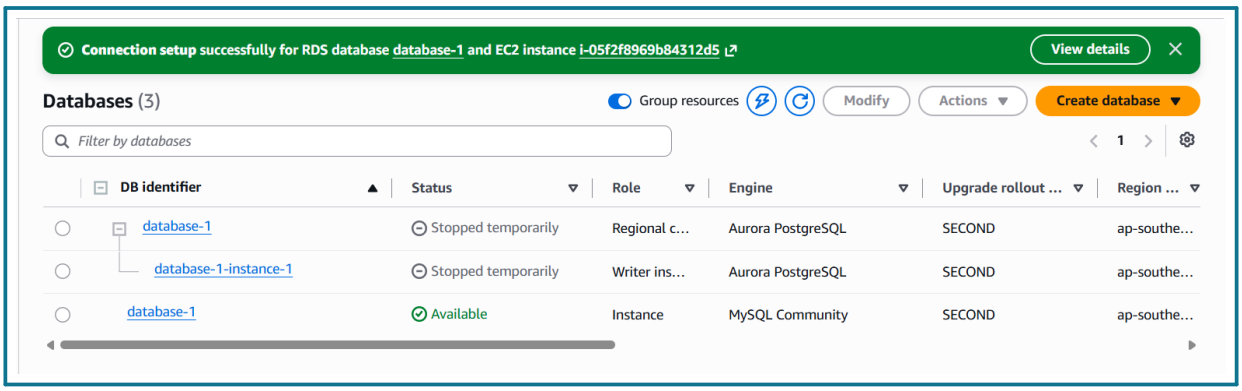


Figure 22 · Connection set up successfully

NOTES

- The Aurora PostgreSQL cluster above it is the older stopped database and is not part of this setup.

PART 4

Steps 23–28

Test the connection from EC2

23 Install the MySQL client on EC2

In the EC2 terminal install the MySQL client so you can talk to the RDS database:

```
$ sudo apt-get install mysql-client -y
```

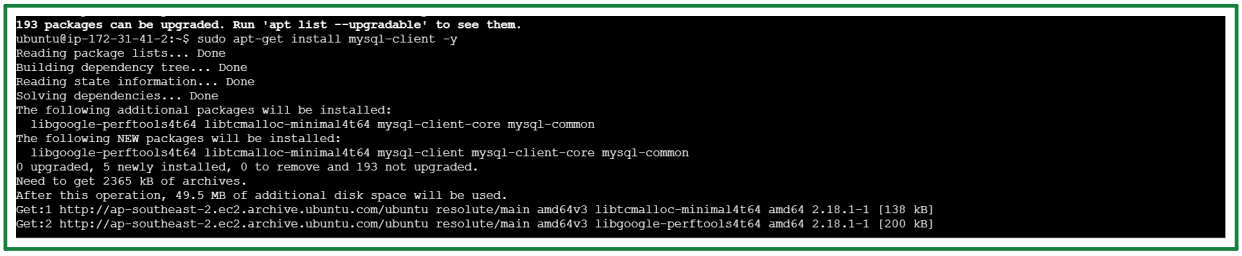


Figure 23 · Install the MySQL client on EC2

NOTES

- The install pulls in mysql-client, mysql-client-core and mysql-common (about 49.5 MB of disk space).

24 Open the database Connectivity & security tab

In RDS, open database-1 and select the Connectivity & security tab. Under Connect using, Endpoints shows the connection details: master username admin and port 3306. Internet access gateway is Disabled.

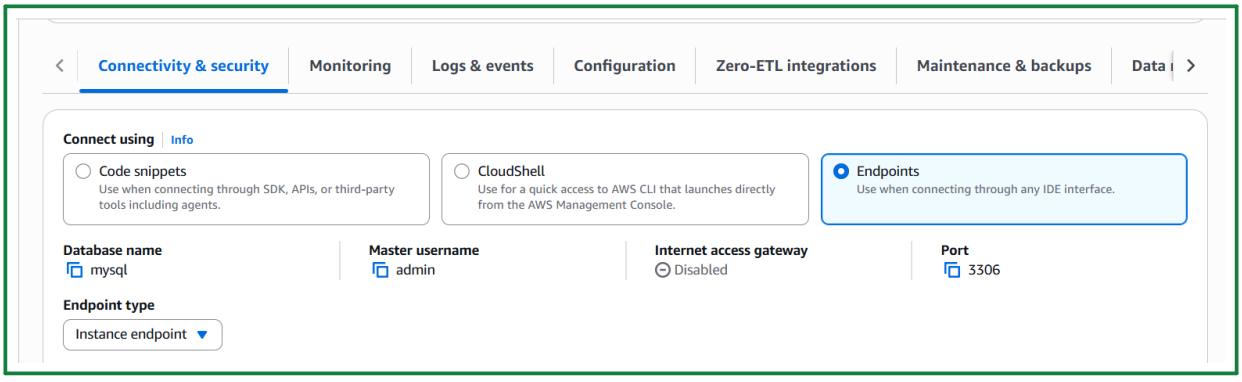


Figure 24 · Open the database Connectivity & security tab

25 Find the database endpoint

The Endpoint & port panel shows the endpoint hostname and port 3306. Networking shows the database is in ap-southeast-2a in the same VPC as the EC2 instance. Security lists the default and rds-ec2-1 security groups (both Active), and Publicly accessible is No.

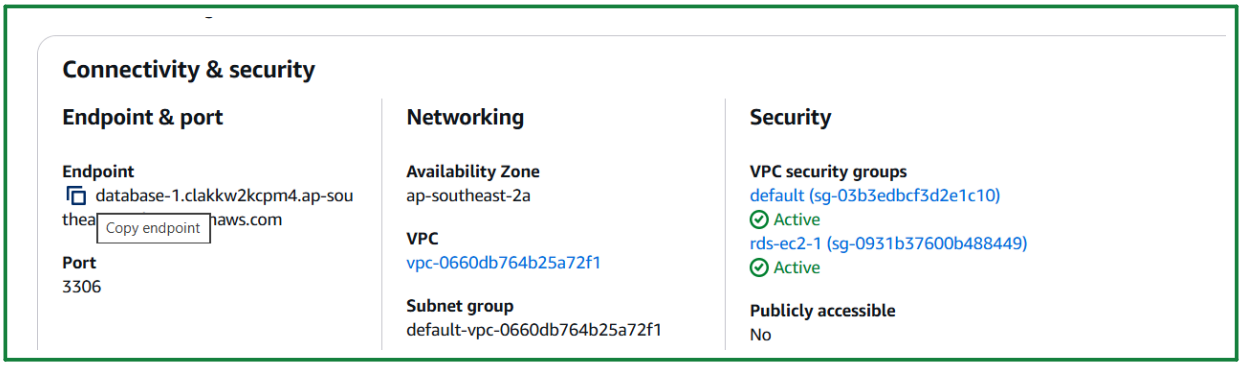


Figure 25 · Find the database endpoint

NOTES

- Publicly accessible = No means the database can only be reached from inside the VPC - this is the secure setup.
- Hover over the endpoint and click the copy icon.

26 Copy the endpoint

A green Endpoint copied message confirms the hostname is on your clipboard. It ends with .ap-southeast-2.rds.amazonaws.com.

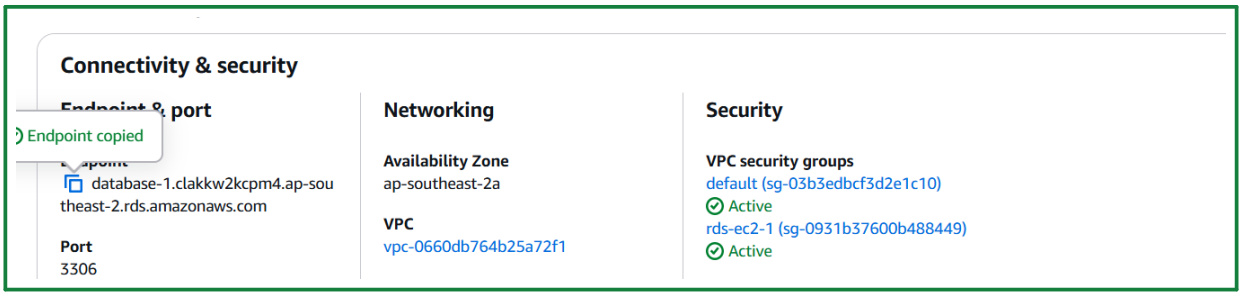


Figure 26 · Copy the endpoint

27 Connect to RDS from the EC2 terminal

Connect using the admin user, the RDS endpoint and port 3306, then enter the master password:

```
$ mysql -u admin -h <RDS-ENDPOINT> -P 3306 -p
```

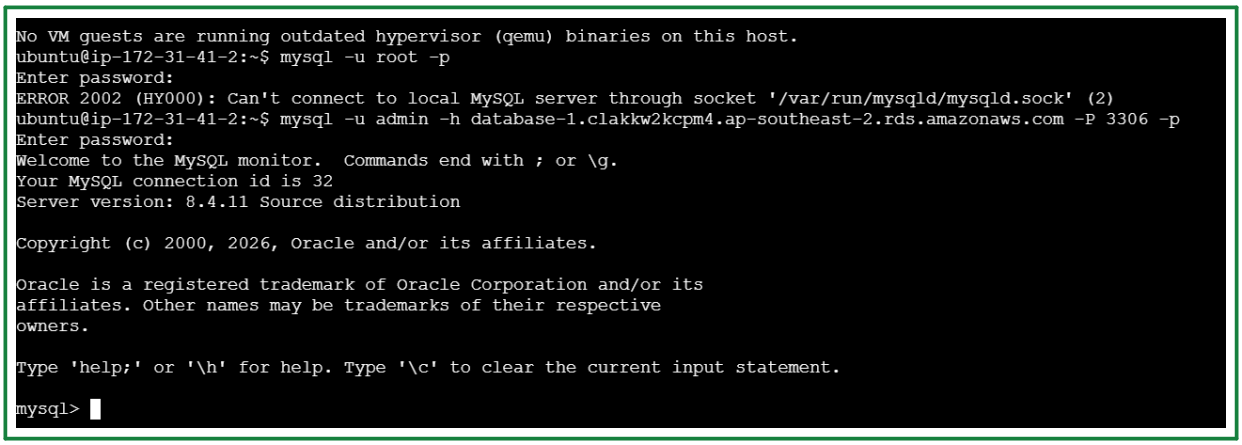


Figure 27 · Connect to RDS from the EC2 terminal

NOTES

- On success the MySQL monitor opens and shows Server version 8.4.11 - the EC2 to RDS connection works.

ATTENTION

- The first attempt, mysql -u root -p, failed with ERROR 2002 (Can't connect to local MySQL server through socket). That command looks for a MySQL server on the EC2 machine itself - always pass -h with the RDS endpoint.

28 Run a test query

Run show databases; and the four default MySQL databases are listed: information_schema, mysql, performance_schema and sys.

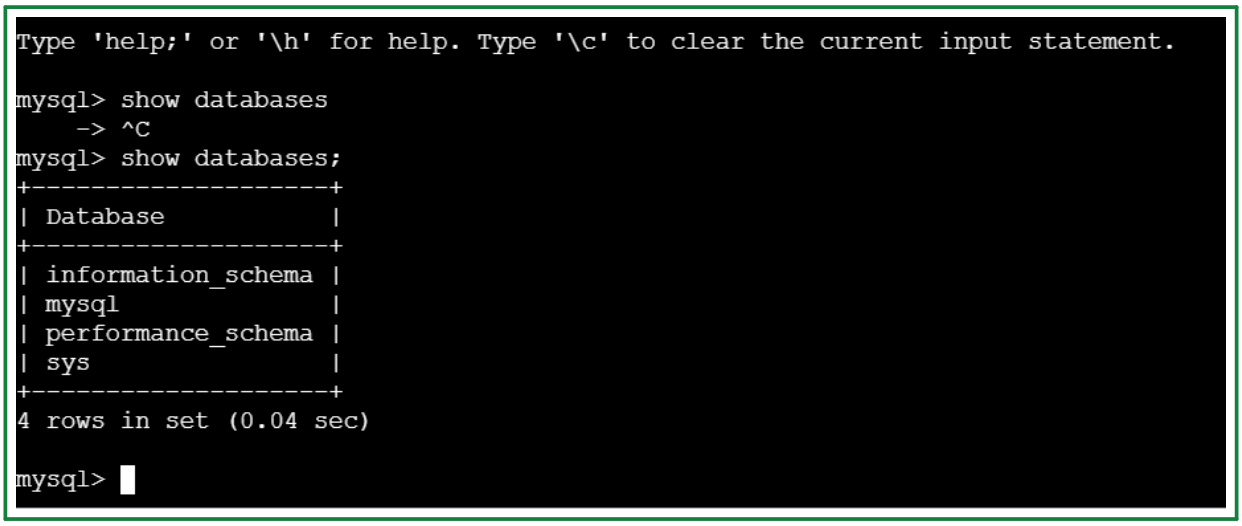


Figure 28 · Run a test query

NOTES

- Every SQL statement must end with a semicolon. Without it the prompt changes to -> and waits for more input (as in the first attempt); press Ctrl+C to cancel and try again.
- From here you can create your own database, for example: CREATE DATABASE testdb;

REFERENCE

Quick command reference

Run these on the EC2 instance (Ubuntu). Replace the placeholder with your own RDS endpoint.

```
$ sudo apt update -y
$ sudo apt-get install mysql-client -y
$ mysql -u admin -h <RDS-ENDPOINT> -P 3306 -p

-- inside the MySQL prompt:
$ show databases;
```

Notes and recommendations

SECURITY	SSH is open to 0.0.0.0/0 on the EC2 security group. Restrict it to your own IP, or remove it and use EC2 Instance Connect.
KEY PAIR	The instance was launched without a key pair. Create one if you want to SSH from your own computer.
CREDENTIALS	Use a strong master password, or choose Managed in AWS Secrets Manager. Never commit passwords to code.
COST	The Free tier has limits. Stop or delete the RDS instance and terminate the EC2 instance when finished to avoid charges.
AVAILABILITY ZONE	The database (ap-southeast-2a) and EC2 instance (ap-southeast-2b) are in different AZs, so cross-AZ data transfer charges may apply. Use the same AZ for both to avoid this.
PRIVACY	The screenshots show the AWS account name, account ID and the full RDS endpoint. Mask them before sharing this document publicly.